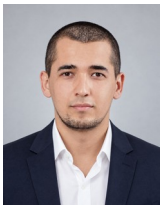
Sean Kakhkhorov



Full-Stack Software Engineer (TypeScript · Next.js · NestJS · FastAPI)

Suwon, South Korea | kakhkhorov97@gmail.com | +82-10-7901-7555
Portfolio: https://www.sean-kakhkhorov.site | GitHub: https://github.com/kakhkhorov97
Work authorization: Bachelor's degree from a Korean university; eligible to change to an E-7 visa upon hiring

PROFESSIONAL SUMMARY

Full-stack software engineer with about 3 years of professional experience building and deploying production web applications, REST/GraphQL APIs, and Linux infrastructure for 20+ client companies, plus 4 end-to-end AI web platforms from data modeling to cloud deployment. Experienced in transactional concurrency control, Toss Payments verification, hybrid RAG search, and CI/CD. Bachelor's degree completed in Korean; fluent English.

TECHNICAL SKILLS

Languages and Frontend: TypeScript, JavaScript, Python, SQL, React, Next.js (App Router), Tailwind CSS, Apollo Client, Redux Toolkit, next-intl

Backend and Data: Node.js, NestJS, Express, FastAPI, GraphQL, REST, Socket.IO, JWT, OAuth 2.0 (PKCE), PostgreSQL, Drizzle ORM, MongoDB, Redis, BullMQ

Applied AI and RAG: Gemini API (structured output, embeddings), hybrid RAG (BM25 + vector, RRF), pgvector, ChromaDB, Kiwi Korean tokenizer, Zod, Pydantic

Infrastructure and QA: Docker, Ubuntu Linux, Nginx, Google Cloud Run, GitHub Actions (CI/CD), Cloudflare, Jest, Vitest, Supertest, pytest (700+ tests)

PROFESSIONAL EXPERIENCE

IT Local — Full-Stack Web Developer | Tashkent · B2B web agency · full-time | Sep 2023 – Apr 2026 (contract ended)

- Core team of 4 developers (about 20 staff). Built an apartment-sales client's admin dashboard (Next.js, React/MUI, GraphQL, MongoDB) with 4 developers, owning **role-based access control (Super Admin / Manager / Sales Staff), the inventory GraphQL API and update mutations, and the admin table UI**. Also built and maintained websites for 20+ clients, including a construction client's systems.
- **Deploy automation:** replaced manual SSH deploys (15–20 min) with GitHub Actions CI/CD (4–5 min, push to live); started on my assigned client projects, then adopted by the team as a template on 5+ sites.
- **Servers:** grew from 5 to 15 Ubuntu servers, provisioning new environments (Nginx reverse proxy, SSL renewal, DNS, PM2, Docker, UFW/SSH hardening, automated backups).
- **Performance:** tens of thousands of catalogue records took about 3–5 s per search because of full collection scans; compound indexes and cursor-based pagination cut it to **under 1 s**.
- **Incidents:** traced an out-of-memory outage (about 15–20 min down) caused by uploads buffered in memory; added upload size/concurrency limits and PM2 auto-restart. Fixed 3–4 major outages; wrote API/deploy docs and a handover.

Golden Valley Investment (formerly Atlantis Investment) — Junior Backend Developer | Tashkent-based real-estate company · remote · project-based contract | Apr 2023 – Sep 2023

- Maintained the company's existing Express.js / MongoDB backend: bug fixes, small feature changes, JWT/session auth and file-upload related code, MongoDB indexes for search and pagination.

PROJECTS

CreatorHub — Multilingual Creator Marketplace | Next.js, NestJS (GraphQL), MongoDB, Socket.IO, Docker, Nginx | Jul 2026 – Present
Live: https://creatorhub.site/ | **Code:** https://github.com/kakhkhorov97/CreatorHub

- EN/KO/UZ marketplace with Socket.IO chat, admin moderation, and a Gemini assistant; transaction-checked request keys and unique indexes prevent duplicate inquiries under concurrency.
- Restart-safe notification worker using 60-second distributed leases and batch checkpoints; Docker/Nginx VPS with GitHub Actions CI and health-checked automatic rollbacks.

Sofiya — AI Book Concierge and E-Commerce Store | Next.js 15, NestJS 11, PostgreSQL (pgvector), Redis, Cloud Run
Live: https://sofiya.store/ | **Code:** https://github.com/kakhkhorov97/Sofiya-Bookstore/tree/v2

- EN/KO/RU concierge fusing pgvector and full-text search via RRF with in-stock filtering and verified citations: 100% Hit@5, 93% Hit@1, 0.95 MRR (vs. 85% Hit@5 for keyword search).
- Toss Payments with server-side amount validation, idempotent checkout, and automatic refunds; keyless OIDC CI/CD to Cloud Run, Google OAuth (PKCE), Redis rate limiting.

Korea Visa Assistant — Bilingual Immigration Law RAG | FastAPI, Python, Kiwi, ChromaDB, BM25, Cloud Run
Live: https://korea-visa-rag-jwq77bpivq-uc.a.run.app/ | **Code:** https://github.com/kakhkhorov97/korea-visa-rag

- EN/KO RAG over 21 visa categories with statute chunking, Kiwi tokenization preserving visa codes, and BM25 + Chroma RRF: 96.9% Hit@5, 0.935 MRR (+33% over 0.702 BM25), with quote-checked citations, raw-statute fallback, and PII blocking.

CargoCheck AI — Trade Document Verification System | BullMQ, Redis, Gemini API, Zod, PostgreSQL, Docker
Live: https://cargocheck-ai-web.vercel.app/ | **Code:** https://github.com/kakhkhorov97/cargocheck-ai

- Cross-verifies invoices and packing lists with BullMQ workers, Zod-validated Gemini extraction, human-in-the-loop review, quantity/unit/decimal checks, PDF audit reports, and file signature checks.

EDUCATION AND LANGUAGES

Chonnam National University (Yeosu, South Korea) — B.A. in International Trade, taught entirely in Korean | Sep 2019 – Aug 2023

Languages: Korean (completed a full bachelor's degree taught in Korean; can handle technical interviews and read technical documents), English (fluent), Uzbek (native), Russian (elementary)